

TEACHER GUIDE • PURPOSE & PACKAGE

Sandbag Emergency Logistics • Zombie Invasion Prep

Central lesson: Emergency response is a logistics problem.

Bags → fill → transportation → people → equipment → time → sustainment.

Having a resource is different from being able to employ it. A strong plan meets the mission with available resources, recognizes bottlenecks, and changes when conditions change.

Print and distribute

Part 1: pages 1–2 per four-person team; reference pages 3–5 per student. Part 2 planning worksheet: three pages per student. Scenario bank: one different one-page card per student; 30 available. Updates: cut the six four-card pages into 24 reusable cards. Teacher guide is separate; do not distribute its estimates before planning.

Editable Markdown files accompany every PDF, plus a complete editable package and the generation script/data.

Print US Letter at actual size; grayscale-friendly design. No weights or online research are needed from students.

Suggested sequence

Session 1: 10–15 min briefing and demonstration, 20 min field trial, 10 min cleanup/count/calculations, 10 min comparison. Reserve additional time for setup, handwashing and travel outside.

Session 2: review common rules (10 min), assign individual cards and plan (30–40 min), introduce one update and revise (10–15 min), debrief (10 min). Use a third session for presentation or extensions if needed.

Teams share their experimental rate, but each student owns a different emergency plan. Distribute cards randomly or by challenge; scenarios 11, 12, 15, 19, 26, 27 and 30 invite comparison, redesign or rejection. No neighborhood or street knowledge is required. All situations are fictional regional composites.

Objectives and evidence

Students measure an actual production chain, calculate rates/fill/team-hours, distinguish ideal and elapsed time, allocate workers and tools, and support workers. Evidence includes field data, labeled sketch, calculations, schedule, resource requests, decision and revised plan. Do not award the highest score merely for the greatest bag count or the largest barrier.

Availability of roughly 50 physical bags

With 30 students you may have seven full teams plus two support observers. Fifty bags total may limit a simultaneous trial. Obtain extra bags when possible, or run staggered trials using the same bags only between trials (teacher-approved emptying/reset). Allocate enough bags that a team is unlikely to run out; monitor and replenish before a queue forms. Record supply-limited trials explicitly. Do not interpret a team cap as its physical maximum.

TEACHER GUIDE • FIELD PROTOCOL & SAFETY

Before students arrive

Inspect a dry, level practice area away from roads, drains and unstable slopes. Use clean sand, flexible bags, appropriately sized shovels, gloves, safety glasses and closed-toe footwear. Mark a short carry route, fill zones and a destination large enough for neat placement. Provide safe drinking water, rest space, a first-aid adult and handwashing. Verify school supervision and weather procedures. Offer meaningful non-lifting roles and accessible alternatives.

Standardize the experiment

Make a sample bag half to two-thirds full and comfortably manageable. This is a visual/tactile standard, not a weighed target. Underfill for safe handling when needed, but note differing standards when comparing teams. Demonstrate consistent closure: fold/tuck or teacher-approved tie. Do not require students to tie densely packed bags. Teach safe shoveling, controlled lifting/placement and buddy handling; no throwing or running. A teacher inspection determines acceptable completion.

Keep four workers, 20 minutes, route distance, fill standard and count definition consistent when possible. Record every equipment difference. Supply gloves and glasses for everyone. Bags counted must be filled, closed, moved and placed at the designated endpoint. Do not count work in progress when the timer ends.

Run the trial

Allow teams to choose roles. Start the clock together; observe queues and role changes. Permit rest and safety stops without penalty. Stop immediately for lightning/thunder, unsafe weather, pain, fatigue or any hazardous condition. Follow school restart policy; never turn a weather stop into a speed challenge. At 20 min, stop all production and count accepted bags once. Record early material exhaustion, elapsed time and route conditions. A teacher may run another supplied trial later, or assign a clearly labeled classroom practice rate if the result is zero/invalid. Example practice rate: B = 20; R = 60. Do not replace genuine observations without labeling them. Do not normalize a capped count as though it measured unconstrained capacity.

Cleanup

Adults control any emptying/reset. Avoid airborne dust; return sand to the designated pile, store bags dry, inspect tools, account for students and wash hands before food. No student-built emergency structure is tested with floodwater.

Planning-scenario boundaries

All field production is on the safe practice site. Cards about floods, electrical equipment, contaminants, slopes or emergency vehicles are paper exercises for adult response planning. Gloves/glasses do not authorize hazardous work. Bags are not watertight, a flood guarantee, a retaining wall or a tent wind rating. Do not imply that a class estimate is a real construction specification. Real deployment needs agency approval, site assessment, proper materials and safe withdrawal decisions.

TEACHER GUIDE • CALCULATION & ASSESSMENT

Use each student's measured R; there is no universal completion time.

Baseline estimates on following pages have no contingency. Accept a reasoned 10% extra or alternative safe layout; require the student to check its resource impact. Basin estimates omit outer-row/corner adjustments deliberately. Cards give mission dimensions. Students select and justify a classroom counting rule; there is no assigned structure or resource stock.

Worked method, not a fixed answer key

If B = 20 bags, R = 60 bags/team-hour. Normal S = 45; difficult S = 30. A 120-bag mission needs 2 experimental team-hours or 2.67 adjusted team-hours at normal conditions. Two fully equipped production teams would take at least 1.33 h. That is not a guaranteed finish: staffing transport separately, delivery, route, setup and inspection can increase elapsed time.

The catalog's starting point is two shovels per team. Students must request enough tools for their chosen workforce and explain any different allocation. Never credit three full-rate teams solely because 12 people are present. Supplementary drivers/operators must be counted. A two-worker support group cannot be treated as a measured four-worker team.

At 1 wheelbarrow, 4 bags/10 min gives a catalog upper bound of 24 bags/h; 1 cart gives 48 bags/h. On 301–600 ft routes these fall to 12 and 24 bags/h. These are movement-only ceilings with an operator, not added guaranteed production. Pickups move at most 80 bags/h under the supplied cycle. Shared loading labor reduces production capacity. On steep routes catalog cycles are optimistic; seek a new route or lower capacity rather than treating 0.50 as a precise transport model.

Do not add normal trial carrying time again. On changed routes, compare the chain's limiting rate and show a staffing/schedule method. Minimum deadline teams is an ideal lower bound, not a promise of feasibility. Give credit for identifying unquantified setup and adding a labeled allowance (e.g., 15–30 min); don't demand a hidden standard.

Sustainment example

Eight workers for 3 elapsed hours: $8 \times 3 \div 4 = 6$ gal base; with 25% reserve, round up 7.5 to 8 gal. Two 5 gal coolers, with accessible refill, provide adequate capacity. Count support staff too. More than 4 h: food servings = workers $\times$ round up(elapsed hours $\div$ 4); normal breaks remain in the 0.75 rate and should not be added a second time unless extra meal breaks are chosen.

Suggested assessment • 20 points

Field evidence and honest rate labeling: 3. Sketch/bag/fill reasoning: 4. People/tools/movement/schedule feasibility: 5. Sustainment/safety/materials: 3. Practical decision and alternatives: 3. Incident revision with changed priorities/resources: 2.

Accept justified ranges and uncertainty. Major omissions include treating empty bags as completed protection, ignoring sand delivery, double-counting workers, claiming unlimited scaling, ignoring access, or continuing a fundamentally unsafe mission. A well-supported rejection can earn full credit.

TEACHER GUIDE • REVISED REQUEST CARDS

The revised scenario cards intentionally omit supplied workers, equipment and materials, along with the long planning-question block. Students determine quantities instead of working backward from a fixed allocation. Counting rules remain in the separate common reference.

Required comparison

For chosen bag count N and sustained rate S , estimate one-team time N/S and four-team time $N/(4S)$. These are ideal production-chain estimates. Then determine teams = round up $[N/(S \times \text{usable hours})]$ to meet the request. Usable hours excludes unavoidable delivery delay and separate setup/finish allowance. Multiply production teams by four and add any separately staffed support roles.

Request shovels (starting point: two per team), gloves/glasses for every worker, fill, transport, other materials and worker support. Test whether the movement chain and workspace can support the chosen team count. Additional teams do not remove the delivery, space or safety constraint.

Judgment and alternatives

The following estimates illustrate reasonable classroom choices; they do not prescribe a structure. Lower narrow lines may suit shallow diversion, while a broad barrier may suit shallow standing water. Students must explain the choice and any changed mission. A narrow line is not automatically an adequate substitute for flood protection. Equipment station counts remain on relevant cards because they are approved mission requirements, not supplies. No bag shortages are imposed in the initial cards. Resource limits can be introduced with incident updates against each student's own requested allocation. When an update refers to listed or available workers/equipment, apply it to the student's proposed resources. A safe redesign or rejection can earn full credit.

Deadline choices

Time windows range from 1 to 12 hours. A large request can require many teams; accept 12 four-person teams if the measured rate and feasible movement/support plan justify them. Do not cap staffing at the previous card allocations. Unsafe flood, slope, electrical or contaminated conditions cannot be solved solely by adding workers.

ESTIMATE GUIDE 01 • Hillside Library Entrance

Possible approach, not a prescribed solution

A narrow two-layer diversion line is one classroom option for shallow runoff. Justify the route and receiving drainage; choose another count if the mission/layout changes.

Illustrative estimate

48 bags; fill = $48/75 = 0.64 \text{ yd}^3$; quarter-yard order 0.75 yd^3 . Allow justified return/corner or contingency changes. Student must show the actual selected rule and dimensions.

One team, four teams and deadline staffing

$S = 0.75R$ normally, or $0.50R$ for justified difficult conditions.

One-team time: $48/S$ hours; four-team time: $48/(4S)$ hours.

Adjusted team-hours: $48/S$. Deadline: 2 hours.

Minimum production teams: round up $[48/(S \times \text{usable hours})]$. Delivery delay/route constraints: A 6 ft accessible entrance must remain open; do not send water toward neighboring property.

For illustration only, if $R = 60$ then normal $S = 45$: one team 1.07 h; four teams 0.27 h. Deadline lower bound ignoring setup/delivery is 1 teams. Substitute the student's measured R and actual usable time.

Equipment and actual elapsed time

Start with 2 shovels per production team and one gloves/glasses set per worker, including support workers. Request fill delivery, appropriate moving equipment, closure tools, first aid, water and other materials. Check the route against trial distance.

On a dry level route ≤ 300 ft, each separately staffed wheelbarrow moves at most 24 bags/h and each cart 48 bags/h under catalog assumptions; longer or difficult routes are slower. Round trips up and include operators/loading help. Vehicle access is not guaranteed. Equipment requests must match the chosen number of teams rather than assumed fixed card stock.

Major constraints and judgment

Reasonable for shallow runoff only. Verify the receiving swale remains safe. Plastic apron, cones, accessible route and runoff observation.

Students should identify the likely slowest stage, actual completion time, safe access/drainage, material dimensions, drinking-water delivery, rest/rotation and food for longer work. Accept fewer bags only when a reduced mission is explicit; accept more teams only with equipment, safe workspace and support to use them.

ESTIMATE GUIDE 02 • Creekside Community Center

Possible approach, not a prescribed solution

A broad three-layer shallow barrier is one classroom option, with returns included or separately estimated. Check forecast depth and leakage. Card 12 is unsafe as a road defense regardless of staffing.

Illustrative estimate

216 bags; fill = $216/75 = 2.88 \text{ yd}^3$; quarter-yard order 3.00 yd^3 . Allow justified return/corner or contingency changes. Student must show the actual selected rule and dimensions.

One team, four teams and deadline staffing

$S = 0.75R$ normally, or $0.50R$ for justified difficult conditions.

One-team time: $216/S$ hours; four-team time: $216/(4S)$ hours.

Adjusted team-hours: $216/S$. Deadline: 4 hours.

Minimum production teams: round up $[216/(S \times \text{usable hours})]$. Delivery delay/route constraints: Protect the occupied building while preserving exits and a way for rainwater behind the barrier to drain.

For illustration only, if $R = 60$ then normal $S = 45$: one team 4.80 h; four teams 1.20 h. Deadline lower bound ignoring setup/delivery is 2 teams. Substitute the student's measured R and actual usable time.

Equipment and actual elapsed time

Start with 2 shovels per production team and one gloves/glasses set per worker, including support workers. Request fill delivery, appropriate moving equipment, closure tools, first aid, water and other materials. Check the route against trial distance.

On a dry level route ≤ 300 ft, each separately staffed wheelbarrow moves at most 24 bags/h and each cart 48 bags/h under catalog assumptions; longer or difficult routes are slower. Round trips up and include operators/loading help. Vehicle access is not guaranteed. Equipment requests must match the chosen number of teams rather than assumed fixed card stock.

Major constraints and judgment

Conditional: shallow, slow water only; leakage and drainage still matter. Plastic overlap, drainage plan, relocation trigger.

Students should identify the likely slowest stage, actual completion time, safe access/drainage, material dimensions, drinking-water delivery, rest/rotation and food for longer work. Accept fewer bags only when a reduced mission is explicit; accept more teams only with equipment, safe workspace and support to use them.

ESTIMATE GUIDE 03 • Shelter Accessible Entry

Possible approach, not a prescribed solution

A narrow two-layer diversion line is one classroom option for shallow runoff. Justify the route and receiving drainage; choose another count if the mission/layout changes.

Illustrative estimate

72 bags; fill = $72/75 = 0.96 \text{ yd}^3$; quarter-yard order 1.00 yd^3 . Allow justified return/corner or contingency changes. Student must show the actual selected rule and dimensions.

One team, four teams and deadline staffing

$S = 0.75R$ normally, or $0.50R$ for justified difficult conditions.

One-team time: $72/S$ hours; four-team time: $72/(4S)$ hours.

Adjusted team-hours: $72/S$. Deadline: 2 hours.

Minimum production teams: round up $[72/(S \times \text{usable hours})]$. Delivery delay/route constraints: Wheelchairs, families and emergency staff need continuous access.

For illustration only, if $R = 60$ then normal $S = 45$: one team 1.60 h; four teams 0.40 h. Deadline lower bound ignoring setup/delivery is 1 teams. Substitute the student's measured R and actual usable time.

Equipment and actual elapsed time

Start with 2 shovels per production team and one gloves/glasses set per worker, including support workers. Request fill delivery, appropriate moving equipment, closure tools, first aid, water and other materials. Check the route against trial distance.

On a dry level route ≤ 300 ft, each separately staffed wheelbarrow moves at most 24 bags/h and each cart 48 bags/h under catalog assumptions; longer or difficult routes are slower. Round trips up and include operators/loading help. Vehicle access is not guaranteed. Equipment requests must match the chosen number of teams rather than assumed fixed card stock.

Major constraints and judgment

Reasonable if approved drainage carries water away without crossing the entrance. Accessible detour during work, signs, plastic apron.

Students should identify the likely slowest stage, actual completion time, safe access/drainage, material dimensions, drinking-water delivery, rest/rotation and food for longer work. Accept fewer bags only when a reduced mission is explicit; accept more teams only with equipment, safe workspace and support to use them.

ESTIMATE GUIDE 04 • Communications Trailer

Possible approach, not a prescribed solution

A narrow two-layer diversion line is one classroom option for shallow runoff. Justify the route and receiving drainage; choose another count if the mission/layout changes.

Illustrative estimate

64 bags; fill = $64/75 = 0.85 \text{ yd}^3$; quarter-yard order 1.00 yd^3 . Allow justified return/corner or contingency changes. Student must show the actual selected rule and dimensions.

One team, four teams and deadline staffing

$S = 0.75R$ normally, or $0.50R$ for justified difficult conditions.

One-team time: $64/S$ hours; four-team time: $64/(4S)$ hours.

Adjusted team-hours: $64/S$. Deadline: 2 hours.

Minimum production teams: round up $[64/(S \times \text{usable hours})]$. Delivery delay/route constraints: Communications must stay operational. Vents, cables, access and a safe drainage outlet must remain clear.

For illustration only, if $R = 60$ then normal $S = 45$: one team 1.42 h; four teams 0.36 h. Deadline lower bound ignoring setup/delivery is 1 teams. Substitute the student's measured R and actual usable time.

Equipment and actual elapsed time

Start with 2 shovels per production team and one gloves/glasses set per worker, including support workers. Request fill delivery, appropriate moving equipment, closure tools, first aid, water and other materials. Check the route against trial distance.

On a dry level route ≤ 300 ft, each separately staffed wheelbarrow moves at most 24 bags/h and each cart 48 bags/h under catalog assumptions; longer or difficult routes are slower. Round trips up and include operators/loading help. Vehicle access is not guaranteed. Equipment requests must match the chosen number of teams rather than assumed fixed card stock.

Major constraints and judgment

Useful as partial protection; move the trailer if the site starts flooding. Cable protection, plastic apron, equipment relocation plan.

Students should identify the likely slowest stage, actual completion time, safe access/drainage, material dimensions, drinking-water delivery, rest/rotation and food for longer work. Accept fewer bags only when a reduced mission is explicit; accept more teams only with equipment, safe workspace and support to use them.

ESTIMATE GUIDE 05 • Utility Cabinet on a Hill

Possible approach, not a prescribed solution

A narrow two-layer diversion line is one classroom option for shallow runoff. Justify the route and receiving drainage; choose another count if the mission/layout changes.

Illustrative estimate

40 bags; fill = $40/75 = 0.53 \text{ yd}^3$; quarter-yard order 0.75 yd^3 . Allow justified return/corner or contingency changes. Student must show the actual selected rule and dimensions.

One team, four teams and deadline staffing

$S = 0.75R$ normally, or $0.50R$ for justified difficult conditions.

One-team time: $40/S$ hours; four-team time: $40/(4S)$ hours.

Adjusted team-hours: $40/S$. Deadline: 1 hours.

Minimum production teams: round up $[40/(S \times \text{usable hours})]$. Delivery delay/route constraints: Utility staff control the exclusion zone and all electrical work. Work must stop if the dry area becomes unsafe.

For illustration only, if $R = 60$ then normal $S = 45$: one team 0.89 h; four teams 0.22 h. Deadline lower bound ignoring setup/delivery is 1 teams. Substitute the student's measured R and actual usable time.

Equipment and actual elapsed time

Start with 2 shovels per production team and one gloves/glasses set per worker, including support workers. Request fill delivery, appropriate moving equipment, closure tools, first aid, water and other materials. Check the route against trial distance.

On a dry level route ≤ 300 ft, each separately staffed wheelbarrow moves at most 24 bags/h and each cart 48 bags/h under catalog assumptions; longer or difficult routes are slower. Round trips up and include operators/loading help. Vehicle access is not guaranteed. Equipment requests must match the chosen number of teams rather than assumed fixed card stock.

Major constraints and judgment

Conditional: utility approval and isolation first; retreat if water enters exclusion zone. Utility coordination, cones, plastic, relocation or isolation.

Students should identify the likely slowest stage, actual completion time, safe access/drainage, material dimensions, drinking-water delivery, rest/rotation and food for longer work. Accept fewer bags only when a reduced mission is explicit; accept more teams only with equipment, safe workspace and support to use them.

ESTIMATE GUIDE 06 • Fire Station Apron

Possible approach, not a prescribed solution

A narrow two-layer diversion line is one classroom option for shallow runoff. Justify the route and receiving drainage; choose another count if the mission/layout changes.

Illustrative estimate

96 bags; fill = $96/75 = 1.28 \text{ yd}^3$; quarter-yard order 1.50 yd^3 . Allow justified return/corner or contingency changes. Student must show the actual selected rule and dimensions.

One team, four teams and deadline staffing

$S = 0.75R$ normally, or $0.50R$ for justified difficult conditions.

One-team time: $96/S$ hours; four-team time: $96/(4S)$ hours.

Adjusted team-hours: $96/S$. Deadline: 3 hours.

Minimum production teams: round up $[96/(S \times \text{usable hours})]$. Delivery delay/route constraints: A 14 ft vehicle exit must remain open. Do not block drainage openings.

For illustration only, if $R = 60$ then normal $S = 45$: one team 2.13 h; four teams 0.53 h. Deadline lower bound ignoring setup/delivery is 1 teams. Substitute the student's measured R and actual usable time.

Equipment and actual elapsed time

Start with 2 shovels per production team and one gloves/glasses set per worker, including support workers. Request fill delivery, appropriate moving equipment, closure tools, first aid, water and other materials. Check the route against trial distance.

On a dry level route ≤ 300 ft, each separately staffed wheelbarrow moves at most 24 bags/h and each cart 48 bags/h under catalog assumptions; longer or difficult routes are slower. Round trips up and include operators/loading help. Vehicle access is not guaranteed. Equipment requests must match the chosen number of teams rather than assumed fixed card stock.

Major constraints and judgment

Reasonable only if public works confirms inlet capacity and destination. Traffic control, plastic, protected inlet and apparatus access.

Students should identify the likely slowest stage, actual completion time, safe access/drainage, material dimensions, drinking-water delivery, rest/rotation and food for longer work. Accept fewer bags only when a reduced mission is explicit; accept more teams only with equipment, safe workspace and support to use them.

ESTIMATE GUIDE 07 • Broken Drain at a School

Possible approach, not a prescribed solution

A narrow two-layer diversion line is one classroom option for shallow runoff. Justify the route and receiving drainage; choose another count if the mission/layout changes.

Illustrative estimate

120 bags; fill = $120/75 = 1.60 \text{ yd}^3$; quarter-yard order 1.75 yd^3 . Allow justified return/corner or contingency changes. Student must show the actual selected rule and dimensions.

One team, four teams and deadline staffing

$S = 0.75R$ normally, or $0.50R$ for justified difficult conditions.

One-team time: $120/S$ hours; four-team time: $120/(4S)$ hours.

Adjusted team-hours: $120/S$. Deadline: 4 hours.

Minimum production teams: round up $[120/(S \times \text{usable hours})]$. Delivery delay/route constraints: The school stays occupied. Keep work zones separated from children and verify the receiving drainage capacity.

For illustration only, if $R = 60$ then normal $S = 45$: one team 2.67 h; four teams 0.67 h. Deadline lower bound ignoring setup/delivery is 1 teams. Substitute the student's measured R and actual usable time.

Equipment and actual elapsed time

Start with 2 shovels per production team and one gloves/glasses set per worker, including support workers. Request fill delivery, appropriate moving equipment, closure tools, first aid, water and other materials. Check the route against trial distance.

On a dry level route ≤ 300 ft, each separately staffed wheelbarrow moves at most 24 bags/h and each cart 48 bags/h under catalog assumptions; longer or difficult routes are slower. Round trips up and include operators/loading help. Vehicle access is not guaranteed. Equipment requests must match the chosen number of teams rather than assumed fixed card stock.

Major constraints and judgment

Useful temporary diversion; more people cannot solve shovel and cart limits. Fencing/cones, plastic, public works repair request.

Students should identify the likely slowest stage, actual completion time, safe access/drainage, material dimensions, drinking-water delivery, rest/rotation and food for longer work. Accept fewer bags only when a reduced mission is explicit; accept more teams only with equipment, safe workspace and support to use them.

ESTIMATE GUIDE 08 • Water-Main Break

Possible approach, not a prescribed solution

A narrow two-layer diversion line is one classroom option for shallow runoff. Justify the route and receiving drainage; choose another count if the mission/layout changes.

Illustrative estimate

60 bags; fill = $60/75 = 0.80 \text{ yd}^3$; quarter-yard order 1.00 yd^3 . Allow justified return/corner or contingency changes. Student must show the actual selected rule and dimensions.

One team, four teams and deadline staffing

$S = 0.75R$ normally, or $0.50R$ for justified difficult conditions.

One-team time: $60/S$ hours; four-team time: $60/(4S)$ hours.

Adjusted team-hours: $60/S$. Deadline: 1.5 hours.

Minimum production teams: round up $[60/(S \times \text{usable hours})]$. Delivery delay/route constraints: Avoid undermined pavement. Compare temporary bagging with source control and relocating supplies.

For illustration only, if $R = 60$ then normal $S = 45$: one team 1.33 h; four teams 0.33 h. Deadline lower bound ignoring setup/delivery is 1 teams. Substitute the student's measured R and actual usable time.

Equipment and actual elapsed time

Start with 2 shovels per production team and one gloves/glasses set per worker, including support workers. Request fill delivery, appropriate moving equipment, closure tools, first aid, water and other materials. Check the route against trial distance.

On a dry level route ≤ 300 ft, each separately staffed wheelbarrow moves at most 24 bags/h and each cart 48 bags/h under catalog assumptions; longer or difficult routes are slower. Round trips up and include operators/loading help. Vehicle access is not guaranteed. Equipment requests must match the chosen number of teams rather than assumed fixed card stock.

Major constraints and judgment

Only a temporary supplement to shutoff, exclusion and professional inspection. Utility contact, cones, plastic, supply relocation.

Students should identify the likely slowest stage, actual completion time, safe access/drainage, material dimensions, drinking-water delivery, rest/rotation and food for longer work. Accept fewer bags only when a reduced mission is explicit; accept more teams only with equipment, safe workspace and support to use them.

ESTIMATE GUIDE 09 • EOC Generator Pad

Possible approach, not a prescribed solution

A narrow two-layer diversion line is one classroom option for shallow runoff. Justify the route and receiving drainage; choose another count if the mission/layout changes.

Illustrative estimate

52 bags; fill = $52/75 = 0.69 \text{ yd}^3$; quarter-yard order 0.75 yd^3 . Allow justified return/corner or contingency changes. Student must show the actual selected rule and dimensions.

One team, four teams and deadline staffing

$S = 0.75R$ normally, or $0.50R$ for justified difficult conditions.

One-team time: $52/S$ hours; four-team time: $52/(4S)$ hours.

Adjusted team-hours: $52/S$. Deadline: 2 hours.

Minimum production teams: round up $[52/(S \times \text{usable hours})]$. Delivery delay/route constraints: The operator must retain fueling access, ventilation and electrical clearances. Relocation may be necessary.

For illustration only, if $R = 60$ then normal $S = 45$: one team 1.16 h; four teams 0.29 h. Deadline lower bound ignoring setup/delivery is 1 teams. Substitute the student's measured R and actual usable time.

Equipment and actual elapsed time

Start with 2 shovels per production team and one gloves/glasses set per worker, including support workers. Request fill delivery, appropriate moving equipment, closure tools, first aid, water and other materials. Check the route against trial distance.

On a dry level route ≤ 300 ft, each separately staffed wheelbarrow moves at most 24 bags/h and each cart 48 bags/h under catalog assumptions; longer or difficult routes are slower. Round trips up and include operators/loading help. Vehicle access is not guaranteed. Equipment requests must match the chosen number of teams rather than assumed fixed card stock.

Major constraints and judgment

Useful only as approved runoff diversion; unsafe electrical conditions stop work. Operator approval, plastic apron, isolation and relocation options.

Students should identify the likely slowest stage, actual completion time, safe access/drainage, material dimensions, drinking-water delivery, rest/rotation and food for longer work. Accept fewer bags only when a reduced mission is explicit; accept more teams only with equipment, safe workspace and support to use them.

ESTIMATE GUIDE 10 • Basement Window Wells

Possible approach, not a prescribed solution

A broad three-layer shallow barrier is one classroom option, with returns included or separately estimated. Check forecast depth and leakage. Card 12 is unsafe as a road defense regardless of staffing.

Illustrative estimate

144 bags; fill = $144/75 = 1.92 \text{ yd}^3$; quarter-yard order 2.00 yd^3 . Allow justified return/corner or contingency changes. Student must show the actual selected rule and dimensions.

One team, four teams and deadline staffing

$S = 0.75R$ normally, or $0.50R$ for justified difficult conditions.

One-team time: $144/S$ hours; four-team time: $144/(4S)$ hours.

Adjusted team-hours: $144/S$. Deadline: 3 hours.

Minimum production teams: round up $[144/(S \times \text{usable hours})]$. Delivery delay/route constraints: Keep emergency exits and ventilation usable; surface protection does not eliminate basement seepage.

For illustration only, if $R = 60$ then normal $S = 45$: one team 3.20 h; four teams 0.80 h. Deadline lower bound ignoring setup/delivery is 2 teams. Substitute the student's measured R and actual usable time.

Equipment and actual elapsed time

Start with 2 shovels per production team and one gloves/glasses set per worker, including support workers. Request fill delivery, appropriate moving equipment, closure tools, first aid, water and other materials. Check the route against trial distance.

On a dry level route ≤ 300 ft, each separately staffed wheelbarrow moves at most 24 bags/h and each cart 48 bags/h under catalog assumptions; longer or difficult routes are slower. Round trips up and include operators/loading help. Vehicle access is not guaranteed. Equipment requests must match the chosen number of teams rather than assumed fixed card stock.

Major constraints and judgment

Targeted protection can be reasonable; seepage and internal water still require monitoring. Plastic, building inspection, drainage and safe pumping by adults.

Students should identify the likely slowest stage, actual completion time, safe access/drainage, material dimensions, drinking-water delivery, rest/rotation and food for longer work. Accept fewer bags only when a reduced mission is explicit; accept more teams only with equipment, safe workspace and support to use them.

ESTIMATE GUIDE 11 • Long Riverfront Frontage

Possible approach, not a prescribed solution

A broad three-layer shallow barrier is one classroom option, with returns included or separately estimated. Check forecast depth and leakage. Card 12 is unsafe as a road defense regardless of staffing.

Illustrative estimate

2,880 bags; fill = $2880/75 = 38.40 \text{ yd}^3$; quarter-yard order 38.50 yd^3 . Allow justified return/corner or contingency changes. Student must show the actual selected rule and dimensions.

One team, four teams and deadline staffing

$S = 0.75R$ normally, or $0.50R$ for justified difficult conditions.

One-team time: $2880/S$ hours; four-team time: $2880/(4S)$ hours.

Adjusted team-hours: $2880/S$. Deadline: 12 hours.

Minimum production teams: round up $[2880/(S \times \text{usable hours})]$. Delivery delay/route constraints: Critical inventory can instead be relocated by adult crews. Compare bagging the whole frontage with targeted protection or relocation. For illustration only, if $R = 60$ then normal $S = 45$: one team 64.00 h; four teams 16.00 h. Deadline lower bound ignoring setup/delivery is 6 teams. Substitute the student's measured R and actual usable time.

Equipment and actual elapsed time

Start with 2 shovels per production team and one gloves/glasses set per worker, including support workers. Request fill delivery, appropriate moving equipment, closure tools, first aid, water and other materials. Check the route against trial distance.

On a dry level route ≤ 300 ft, each separately staffed wheelbarrow moves at most 24 bags/h and each cart 48 bags/h under catalog assumptions; longer or difficult routes are slower. Round trips up and include operators/loading help. Vehicle access is not guaranteed. Equipment requests must match the chosen number of teams rather than assumed fixed card stock.

Major constraints and judgment

Wrong primary plan at this scale: bag shortage, sheet shortage and deadline favor relocating inventory. Inventory priority list, evacuation/relocation, targeted protection.

Students should identify the likely slowest stage, actual completion time, safe access/drainage, material dimensions, drinking-water delivery, rest/rotation and food for longer work. Accept fewer bags only when a reduced mission is explicit; accept more teams only with equipment, safe workspace and support to use them.

ESTIMATE GUIDE 12 • Low Road Crossing

Possible approach, not a prescribed solution

A broad three-layer shallow barrier is one classroom option, with returns included or separately estimated. Check forecast depth and leakage. Card 12 is unsafe as a road defense regardless of staffing.

Illustrative estimate

1,200 bags; fill = $1200/75 = 16.00 \text{ yd}^3$; quarter-yard order 16.00 yd^3 . Allow justified return/corner or contingency changes. Student must show the actual selected rule and dimensions.

One team, four teams and deadline staffing

$S = 0.75R$ normally, or $0.50R$ for justified difficult conditions.

One-team time: $1200/S$ hours; four-team time: $1200/(4S)$ hours.

Adjusted team-hours: $1200/S$. Deadline: 2 hours.

Minimum production teams: round up $[1200/(S \times \text{usable hours})]$. Delivery delay/route constraints: A dry detour exists.

Nobody enters moving floodwater; the classroom barrier model is not a roadway flood defense design.

For illustration only, if $R = 60$ then normal $S = 45$: one team 26.67 h; four teams 6.67 h. Deadline lower bound ignoring setup/delivery is 14 teams. Substitute the student's measured R and actual usable time.

Equipment and actual elapsed time

Start with 2 shovels per production team and one gloves/glasses set per worker, including support workers. Request fill delivery, appropriate moving equipment, closure tools, first aid, water and other materials. Check the route against trial distance.

On a dry level route ≤ 300 ft, each separately staffed wheelbarrow moves at most 24 bags/h and each cart 48 bags/h under catalog assumptions; longer or difficult routes are slower. Round trips up and include operators/loading help.

Vehicle access is not guaranteed. Equipment requests must match the chosen number of teams rather than assumed fixed card stock.

Major constraints and judgment

Reject: modeled wall lower than water, moving water, roadway hazards and a short deadline. Road closure, dry detour, public works and responder coordination.

Students should identify the likely slowest stage, actual completion time, safe access/drainage, material dimensions, drinking-water delivery, rest/rotation and food for longer work. Accept fewer bags only when a reduced mission is explicit; accept more teams only with equipment, safe workspace and support to use them.

ESTIMATE GUIDE 13 • Park Trail Washout

Possible approach, not a prescribed solution

A small two-wide, two-layer erosion-control line is one counting option for each affected section, subject to qualified approval. It is not slope, road or bank reinforcement.

Illustrative estimate

96 bags; fill = $96/75 = 1.28 \text{ yd}^3$; quarter-yard order 1.50 yd^3 . Allow justified return/corner or contingency changes. Student must show the actual selected rule and dimensions.

One team, four teams and deadline staffing

$S = 0.75R$ normally, or $0.50R$ for justified difficult conditions.

One-team time: $96/S$ hours; four-team time: $96/(4S)$ hours.

Adjusted team-hours: $96/S$. Deadline: 4 hours.

Minimum production teams: round up $[96/(S \times \text{usable hours})]$. Delivery delay/route constraints: The trail remains closed. The proposed treatment cannot serve as a replacement walking surface.

For illustration only, if $R = 60$ then normal $S = 45$: one team 2.13 h; four teams 0.53 h. Deadline lower bound ignoring setup/delivery is 1 teams. Substitute the student's measured R and actual usable time.

Equipment and actual elapsed time

Start with 2 shovels per production team and one gloves/glasses set per worker, including support workers. Request fill delivery, appropriate moving equipment, closure tools, first aid, water and other materials. Check the route against trial distance.

On a dry level route ≤ 300 ft, each separately staffed wheelbarrow moves at most 24 bags/h and each cart 48 bags/h under catalog assumptions; longer or difficult routes are slower. Round trips up and include operators/loading help. Vehicle access is not guaranteed. Equipment requests must match the chosen number of teams rather than assumed fixed card stock.

Major constraints and judgment

Useful temporary runoff control; bags do not rebuild the trail or carry people. Filter fabric, closure signs and permanent repair.

Students should identify the likely slowest stage, actual completion time, safe access/drainage, material dimensions, drinking-water delivery, rest/rotation and food for longer work. Accept fewer bags only when a reduced mission is explicit; accept more teams only with equipment, safe workspace and support to use them.

ESTIMATE GUIDE 14 • Road Shoulder After a Storm

Possible approach, not a prescribed solution

A small two-wide, two-layer erosion-control line is one counting option for each affected section, subject to qualified approval. It is not slope, road or bank reinforcement.

Illustrative estimate

120 bags; fill = $120/75 = 1.60 \text{ yd}^3$; quarter-yard order 1.75 yd^3 . Allow justified return/corner or contingency changes. Student must show the actual selected rule and dimensions.

One team, four teams and deadline staffing

$S = 0.75R$ normally, or $0.50R$ for justified difficult conditions.

One-team time: $120/S$ hours; four-team time: $120/(4S)$ hours.

Adjusted team-hours: $120/S$. Deadline: 4 hours.

Minimum production teams: round up $[120/(S \times \text{usable hours})]$. Delivery delay/route constraints: Bags cannot support traffic or justify reopening the road. Public Works must approve any temporary treatment.

For illustration only, if $R = 60$ then normal $S = 45$: one team 2.67 h; four teams 0.67 h. Deadline lower bound ignoring setup/delivery is 1 teams. Substitute the student's measured R and actual usable time.

Equipment and actual elapsed time

Start with 2 shovels per production team and one gloves/glasses set per worker, including support workers. Request fill delivery, appropriate moving equipment, closure tools, first aid, water and other materials. Check the route against trial distance.

On a dry level route ≤ 300 ft, each separately staffed wheelbarrow moves at most 24 bags/h and each cart 48 bags/h under catalog assumptions; longer or difficult routes are slower. Round trips up and include operators/loading help.

Vehicle access is not guaranteed. Equipment requests must match the chosen number of teams rather than assumed fixed card stock.

Major constraints and judgment

Temporary surface protection only; public works must assess the road before reopening. Filter fabric, barricades, engineering inspection.

Students should identify the likely slowest stage, actual completion time, safe access/drainage, material dimensions, drinking-water delivery, rest/rotation and food for longer work. Accept fewer bags only when a reduced mission is explicit; accept more teams only with equipment, safe workspace and support to use them.

ESTIMATE GUIDE 15 • Landslide at a Hillside Street

Possible approach, not a prescribed solution

A small two-wide, two-layer erosion-control line is one counting option for each affected section, subject to qualified approval. It is not slope, road or bank reinforcement.

Illustrative estimate

400 bags; fill = $400/75 = 5.33 \text{ yd}^3$; quarter-yard order 5.50 yd^3 . Allow justified return/corner or contingency changes. Student must show the actual selected rule and dimensions.

One team, four teams and deadline staffing

$S = 0.75R$ normally, or $0.50R$ for justified difficult conditions.

One-team time: $400/S$ hours; four-team time: $400/(4S)$ hours.

Adjusted team-hours: $400/S$. Deadline: 3 hours.

Minimum production teams: round up $[400/(S \times \text{usable hours})]$. Delivery delay/route constraints: No entry into the slide zone. Distinguish surface runoff control from actual slope stabilization.

For illustration only, if $R = 60$ then normal $S = 45$: one team 8.89 h; four teams 2.22 h. Deadline lower bound ignoring setup/delivery is 3 teams. Substitute the student's measured R and actual usable time.

Equipment and actual elapsed time

Start with 2 shovels per production team and one gloves/glasses set per worker, including support workers. Request fill delivery, appropriate moving equipment, closure tools, first aid, water and other materials. Check the route against trial distance.

On a dry level route ≤ 300 ft, each separately staffed wheelbarrow moves at most 24 bags/h and each cart 48 bags/h under catalog assumptions; longer or difficult routes are slower. Round trips up and include operators/loading help. Vehicle access is not guaranteed. Equipment requests must match the chosen number of teams rather than assumed fixed card stock.

Major constraints and judgment

Reject as slope support. Surface-control estimates do not establish landslide stability. Exclusion, road closure, utility checks and professional slope assessment.

Students should identify the likely slowest stage, actual completion time, safe access/drainage, material dimensions, drinking-water delivery, rest/rotation and food for longer work. Accept fewer bags only when a reduced mission is explicit; accept more teams only with equipment, safe workspace and support to use them.

ESTIMATE GUIDE 16 • Small Culvert Approach

Possible approach, not a prescribed solution

A small two-wide, two-layer erosion-control line is one counting option for each affected section, subject to qualified approval. It is not slope, road or bank reinforcement.

Illustrative estimate

120 bags; fill = $120/75 = 1.60 \text{ yd}^3$; quarter-yard order 1.75 yd^3 . Allow justified return/corner or contingency changes. Student must show the actual selected rule and dimensions.

One team, four teams and deadline staffing

$S = 0.75R$ normally, or $0.50R$ for justified difficult conditions.

One-team time: $120/S$ hours; four-team time: $120/(4S)$ hours.

Adjusted team-hours: $120/S$. Deadline: 4 hours.

Minimum production teams: round up $[120/(S \times \text{usable hours})]$. Delivery delay/route constraints: Keep the culvert opening clear and the approach closed. Avoid backing water upstream.

For illustration only, if $R = 60$ then normal $S = 45$: one team 2.67 h; four teams 0.67 h. Deadline lower bound ignoring setup/delivery is 1 teams. Substitute the student's measured R and actual usable time.

Equipment and actual elapsed time

Start with 2 shovels per production team and one gloves/glasses set per worker, including support workers. Request fill delivery, appropriate moving equipment, closure tools, first aid, water and other materials. Check the route against trial distance.

On a dry level route ≤ 300 ft, each separately staffed wheelbarrow moves at most 24 bags/h and each cart 48 bags/h under catalog assumptions; longer or difficult routes are slower. Round trips up and include operators/loading help.

Vehicle access is not guaranteed. Equipment requests must match the chosen number of teams rather than assumed fixed card stock.

Major constraints and judgment

Partial solution: retain access restrictions and avoid backing water upstream. Filter fabric, exclusion, culvert inspection and repair.

Students should identify the likely slowest stage, actual completion time, safe access/drainage, material dimensions, drinking-water delivery, rest/rotation and food for longer work. Accept fewer bags only when a reduced mission is explicit; accept more teams only with equipment, safe workspace and support to use them.

ESTIMATE GUIDE 17 • Creek Bank Beside a Ballfield

Possible approach, not a prescribed solution

A small two-wide, two-layer erosion-control line is one counting option for each affected section, subject to qualified approval. It is not slope, road or bank reinforcement.

Illustrative estimate

144 bags; fill = $144/75 = 1.92 \text{ yd}^3$; quarter-yard order 2.00 yd^3 . Allow justified return/corner or contingency changes. Student must show the actual selected rule and dimensions.

One team, four teams and deadline staffing

$S = 0.75R$ normally, or $0.50R$ for justified difficult conditions.

One-team time: $144/S$ hours; four-team time: $144/(4S)$ hours.

Adjusted team-hours: $144/S$. Deadline: 4 hours.

Minimum production teams: round up $[144/(S \times \text{usable hours})]$. Delivery delay/route constraints: No work in the creek or at the unstable bank edge. A professional crew must handle permanent stabilization.

For illustration only, if $R = 60$ then normal $S = 45$: one team 3.20 h; four teams 0.80 h. Deadline lower bound ignoring setup/delivery is 1 teams. Substitute the student's measured R and actual usable time.

Equipment and actual elapsed time

Start with 2 shovels per production team and one gloves/glasses set per worker, including support workers. Request fill delivery, appropriate moving equipment, closure tools, first aid, water and other materials. Check the route against trial distance.

On a dry level route ≤ 300 ft, each separately staffed wheelbarrow moves at most 24 bags/h and each cart 48 bags/h under catalog assumptions; longer or difficult routes are slower. Round trips up and include operators/loading help. Vehicle access is not guaranteed. Equipment requests must match the chosen number of teams rather than assumed fixed card stock.

Major constraints and judgment

Conditional temporary runoff control; bank repair requires qualified crews. Filter fabric, closure and professional bank stabilization.

Students should identify the likely slowest stage, actual completion time, safe access/drainage, material dimensions, drinking-water delivery, rest/rotation and food for longer work. Accept fewer bags only when a reduced mission is explicit; accept more teams only with equipment, safe workspace and support to use them.

ESTIMATE GUIDE 18 • Shelter Cleaning-Water Basin

Possible approach, not a prescribed solution

A two-wide, two-layer perimeter is a starting option for a shallow basin. Card 19 uses a hypothetical four-wide/four-layer count to expose why the requested depth is still infeasible. Safer reduced depths or approved containers are valid alternatives.

Illustrative estimate

160 bags; fill = $160/75 = 2.13 \text{ yd}^3$; quarter-yard order 2.25 yd^3 . Allow justified return/corner or contingency changes. Student must show the actual selected rule and dimensions.

One team, four teams and deadline staffing

$S = 0.75R$ normally, or $0.50R$ for justified difficult conditions.

One-team time: $160/S$ hours; four-team time: $160/(4S)$ hours.

Adjusted team-hours: $160/S$. Deadline: 4 hours.

Minimum production teams: round up $[160/(S \times \text{usable hours})]$. Delivery delay/route constraints: Water is NON-POTABLE. Set a sensible operating depth with freeboard and overflow; rainfall supply is uncertain.

For illustration only, if $R = 60$ then normal $S = 45$: one team 3.56 h; four teams 0.89 h. Deadline lower bound ignoring setup/delivery is 1 teams. Substitute the student's measured R and actual usable time.

Equipment and actual elapsed time

Start with 2 shovels per production team and one gloves/glasses set per worker, including support workers. Request fill delivery, appropriate moving equipment, closure tools, first aid, water and other materials. Check the route against trial distance.

On a dry level route ≤ 300 ft, each separately staffed wheelbarrow moves at most 24 bags/h and each cart 48 bags/h under catalog assumptions; longer or difficult routes are slower. Round trips up and include operators/loading help. Vehicle access is not guaranteed. Equipment requests must match the chosen number of teams rather than assumed fixed card stock.

Major constraints and judgment

Conditional: lined, supervised, ground-level non-potable collection; water safety and availability govern. Liner, labels, barriers, overflow route and adult draw-off plan. Requested basin volume at 0.5 ft depth: 48 ft^3 , 359.04 gallons, approximately 2,980 lb of water. Recalculate for a safer selected operating depth; stored water is not guaranteed rainwater supply.

Students should identify the likely slowest stage, actual completion time, safe access/drainage, material dimensions, drinking-water delivery, rest/rotation and food for longer work. Accept fewer bags only when a reduced mission is explicit; accept more teams only with equipment, safe workspace and support to use them.

ESTIMATE GUIDE 19 • Oversized Rainwater Request

Possible approach, not a prescribed solution

A two-wide, two-layer perimeter is a starting option for a shallow basin. Card 19 uses a hypothetical four-wide/four-layer count to expose why the requested depth is still infeasible. Safer reduced depths or approved containers are valid alternatives.

Illustrative estimate

1,600 bags; fill = $1600/75 = 21.33 \text{ yd}^3$; quarter-yard order 21.50 yd^3 . Allow justified return/corner or contingency changes. Student must show the actual selected rule and dimensions.

One team, four teams and deadline staffing

$S = 0.75R$ normally, or $0.50R$ for justified difficult conditions.

One-team time: $1600/S$ hours; four-team time: $1600/(4S)$ hours.

Adjusted team-hours: $1600/S$. Deadline: 12 hours.

Minimum production teams: round up $[1600/(S \times \text{usable hours})]$. Delivery delay/route constraints: There is no approved deep-basin design. Calculate the water load and compare smaller basins or approved containers.

For illustration only, if $R = 60$ then normal $S = 45$: one team 35.56 h; four teams 8.89 h. Deadline lower bound ignoring setup/delivery is 3 teams. Substitute the student's measured R and actual usable time.

Equipment and actual elapsed time

Start with 2 shovels per production team and one gloves/glasses set per worker, including support workers. Request fill delivery, appropriate moving equipment, closure tools, first aid, water and other materials. Check the route against trial distance.

On a dry level route ≤ 300 ft, each separately staffed wheelbarrow moves at most 24 bags/h and each cart 48 bags/h under catalog assumptions; longer or difficult routes are slower. Round trips up and include operators/loading help. Vehicle access is not guaranteed. Equipment requests must match the chosen number of teams rather than assumed fixed card stock.

Major constraints and judgment

Reject: requested depth exceeds berm, undersized liner, enormous water load and insufficient bags. Approved tanks/containers, water delivery, smaller shallow basin with approval. Requested basin volume at 2 ft depth: 1200 ft^3 , 8976.00 gallons, approximately 74,501 lb of water. Recalculate for a safer selected operating depth; stored water is not guaranteed rainwater supply.

Students should identify the likely slowest stage, actual completion time, safe access/drainage, material dimensions, drinking-water delivery, rest/rotation and food for longer work. Accept fewer bags only when a reduced mission is explicit; accept more teams only with equipment, safe workspace and support to use them.

ESTIMATE GUIDE 20 • Clinic Sanitation Collection

Possible approach, not a prescribed solution

A two-wide, two-layer perimeter is a starting option for a shallow basin. Card 19 uses a hypothetical four-wide/four-layer count to expose why the requested depth is still infeasible. Safer reduced depths or approved containers are valid alternatives.

Illustrative estimate

160 bags; fill = $160/75 = 2.13 \text{ yd}^3$; quarter-yard order 2.25 yd^3 . Allow justified return/corner or contingency changes. Student must show the actual selected rule and dimensions.

One team, four teams and deadline staffing

$S = 0.75R$ normally, or $0.50R$ for justified difficult conditions.

One-team time: $160/S$ hours; four-team time: $160/(4S)$ hours.

Adjusted team-hours: $160/S$. Deadline: 3 hours.

Minimum production teams: round up $[160/(S \times \text{usable hours})]$. Delivery delay/route constraints: No vehicle fits the last 300 ft. Water is not for drinking or handwashing; compare collection with delivered containers.

For illustration only, if $R = 60$ then normal $S = 45$: one team 3.56 h; four teams 0.89 h. Deadline lower bound ignoring setup/delivery is 2 teams. Substitute the student's measured R and actual usable time.

Equipment and actual elapsed time

Start with 2 shovels per production team and one gloves/glasses set per worker, including support workers. Request fill delivery, appropriate moving equipment, closure tools, first aid, water and other materials. Check the route against trial distance.

On a dry level route ≤ 300 ft, each separately staffed wheelbarrow moves at most 24 bags/h and each cart 48 bags/h under catalog assumptions; longer or difficult routes are slower. Round trips up and include operators/loading help. Vehicle access is not guaranteed. Equipment requests must match the chosen number of teams rather than assumed fixed card stock.

Major constraints and judgment

Conditional: movement is likely limiting; delivered approved containers may be easier. Liner, signage, barrier, overflow and authorized sanitation-use plan. Requested basin volume at 0.25 ft depth: 25 ft^3 , 187.00 gallons, approximately 1,552 lb of water. Recalculate for a safer selected operating depth; stored water is not guaranteed rainwater supply.

Students should identify the likely slowest stage, actual completion time, safe access/drainage, material dimensions, drinking-water delivery, rest/rotation and food for longer work. Accept fewer bags only when a reduced mission is explicit; accept more teams only with equipment, safe workspace and support to use them.

ESTIMATE GUIDE 21 • Shelter Tarp Floor

Possible approach, not a prescribed solution

The reference ground-cover rule is one bag per 4 ft of perimeter. This is an inventory estimate for light conditions, not a wind rating.

Illustrative estimate

16 bags; fill = $16/75 = 0.21 \text{ yd}^3$; quarter-yard order 0.25 yd^3 . Allow justified return/corner or contingency changes. Student must show the actual selected rule and dimensions.

One team, four teams and deadline staffing

$S = 0.75R$ normally, or $0.50R$ for justified difficult conditions.

One-team time: $16/S$ hours; four-team time: $16/(4S)$ hours.

Adjusted team-hours: $16/S$. Deadline: 1 hours.

Minimum production teams: round up $[16/(S \times \text{usable hours})]$. Delivery delay/route constraints: This is ground cover, not an overhead shelter. Keep edges out of accessible walking routes.

For illustration only, if $R = 60$ then normal $S = 45$: one team 0.36 h; four teams 0.09 h. Deadline lower bound ignoring setup/delivery is 1 teams. Substitute the student's measured R and actual usable time.

Equipment and actual elapsed time

Start with 2 shovels per production team and one gloves/glasses set per worker, including support workers. Request fill delivery, appropriate moving equipment, closure tools, first aid, water and other materials. Check the route against trial distance.

On a dry level route ≤ 300 ft, each separately staffed wheelbarrow moves at most 24 bags/h and each cart 48 bags/h under catalog assumptions; longer or difficult routes are slower. Round trips up and include operators/loading help. Vehicle access is not guaranteed. Equipment requests must match the chosen number of teams rather than assumed fixed card stock.

Major constraints and judgment

Reasonable in light wind on dry ground; do not infer structural anchoring strength. Tarp, edge markings and trip-hazard control.

Students should identify the likely slowest stage, actual completion time, safe access/drainage, material dimensions, drinking-water delivery, rest/rotation and food for longer work. Accept fewer bags only when a reduced mission is explicit; accept more teams only with equipment, safe workspace and support to use them.

ESTIMATE GUIDE 22 • Temporary Privacy Partitions

Possible approach, not a prescribed solution

Use the supervisor-specified approved stations $\times$ bags per station. The equipment still requires rated supports and required anchors.

Illustrative estimate

12 bags; fill = $12/75 = 0.16 \text{ yd}^3$; quarter-yard order 0.25 yd^3 . Allow justified return/corner or contingency changes. Student must show the actual selected rule and dimensions.

One team, four teams and deadline staffing

$S = 0.75R$ normally, or $0.50R$ for justified difficult conditions.

One-team time: $12/S$ hours; four-team time: $12/(4S)$ hours.

Adjusted team-hours: $12/S$. Deadline: 1 hours.

Minimum production teams: round up $[12/(S \times \text{usable hours})]$. Delivery delay/route constraints: Only the rated frame system may be used; keep exits open and protect the floor.

For illustration only, if $R = 60$ then normal $S = 45$: one team 0.27 h; four teams 0.07 h. Deadline lower bound ignoring setup/delivery is 1 teams. Substitute the student's measured R and actual usable time.

Equipment and actual elapsed time

Start with 2 shovels per production team and one gloves/glasses set per worker, including support workers. Request fill delivery, appropriate moving equipment, closure tools, first aid, water and other materials. Check the route against trial distance.

On a dry level route ≤ 300 ft, each separately staffed wheelbarrow moves at most 24 bags/h and each cart 48 bags/h under catalog assumptions; longer or difficult routes are slower. Round trips up and include operators/loading help. Vehicle access is not guaranteed. Equipment requests must match the chosen number of teams rather than assumed fixed card stock.

Major constraints and judgment

Reasonable only with the supplied rated system and supervisor-approved ballast. Rated frames, floor protection, accessible egress and supervisor check.

Students should identify the likely slowest stage, actual completion time, safe access/drainage, material dimensions, drinking-water delivery, rest/rotation and food for longer work. Accept fewer bags only when a reduced mission is explicit; accept more teams only with equipment, safe workspace and support to use them.

ESTIMATE GUIDE 23 • Mobile Charging Tent

Possible approach, not a prescribed solution

Use the supervisor-specified approved stations $\times$ bags per station. The equipment still requires rated supports and required anchors.

Illustrative estimate

32 bags; fill = $32/75 = 0.43 \text{ yd}^3$; quarter-yard order 0.50 yd^3 . Allow justified return/corner or contingency changes. Student must show the actual selected rule and dimensions.

One team, four teams and deadline staffing

$S = 0.75R$ normally, or $0.50R$ for justified difficult conditions.

One-team time: $32/S$ hours; four-team time: $32/(4S)$ hours.

Adjusted team-hours: $32/S$. Deadline: 2 hours.

Minimum production teams: round up $[32/(S \times \text{usable hours})]$. Delivery delay/route constraints: The manufacturer still controls anchoring. Sandbag quantities do not establish a wind rating; plan indoor shelter for thunderstorms.

For illustration only, if $R = 60$ then normal $S = 45$: one team 0.71 h; four teams 0.18 h. Deadline lower bound ignoring setup/delivery is 1 teams. Substitute the student's measured R and actual usable time.

Equipment and actual elapsed time

Start with 2 shovels per production team and one gloves/glasses set per worker, including support workers. Request fill delivery, appropriate moving equipment, closure tools, first aid, water and other materials. Check the route against trial distance.

On a dry level route ≤ 300 ft, each separately staffed wheelbarrow moves at most 24 bags/h and each cart 48 bags/h under catalog assumptions; longer or difficult routes are slower. Round trips up and include operators/loading help. Vehicle access is not guaranteed. Equipment requests must match the chosen number of teams rather than assumed fixed card stock.

Major constraints and judgment

Conditional supplement only. Suspend use and shelter indoors for threatening weather. Rated anchors, cable covers, electrical supervision and weather shutdown.

Students should identify the likely slowest stage, actual completion time, safe access/drainage, material dimensions, drinking-water delivery, rest/rotation and food for longer work. Accept fewer bags only when a reduced mission is explicit; accept more teams only with equipment, safe workspace and support to use them.

ESTIMATE GUIDE 24 • Emergency Wayfinding Signs

Possible approach, not a prescribed solution

Use the supervisor-specified approved stations $\times$ bags per station. The equipment still requires rated supports and required anchors.

Illustrative estimate

20 bags; fill = $20/75 = 0.27 \text{ yd}^3$; quarter-yard order 0.50 yd^3 . Allow justified return/corner or contingency changes. Student must show the actual selected rule and dimensions.

One team, four teams and deadline staffing

$S = 0.75R$ normally, or $0.50R$ for justified difficult conditions.

One-team time: $20/S$ hours; four-team time: $20/(4S)$ hours.

Adjusted team-hours: $20/S$. Deadline: 1.5 hours.

Minimum production teams: round up $[20/(S \times \text{usable hours})]$. Delivery delay/route constraints: The route stays accessible. Allow time for distributed placement and checking visibility.

For illustration only, if $R = 60$ then normal $S = 45$: one team 0.44 h; four teams 0.11 h. Deadline lower bound ignoring setup/delivery is 1 teams. Substitute the student's measured R and actual usable time.

Equipment and actual elapsed time

Start with 2 shovels per production team and one gloves/glasses set per worker, including support workers. Request fill delivery, appropriate moving equipment, closure tools, first aid, water and other materials. Check the route against trial distance.

On a dry level route ≤ 300 ft, each separately staffed wheelbarrow moves at most 24 bags/h and each cart 48 bags/h under catalog assumptions; longer or difficult routes are slower. Round trips up and include operators/loading help. Vehicle access is not guaranteed. Equipment requests must match the chosen number of teams rather than assumed fixed card stock.

Major constraints and judgment

Reasonable with approved stands; distributed placement and visibility may dominate. Reflective signage, rated stands, lights if work continues.

Students should identify the likely slowest stage, actual completion time, safe access/drainage, material dimensions, drinking-water delivery, rest/rotation and food for longer work. Accept fewer bags only when a reduced mission is explicit; accept more teams only with equipment, safe workspace and support to use them.

ESTIMATE GUIDE 25 • Debris-Sorting Tarp

Possible approach, not a prescribed solution

The reference ground-cover rule is one bag per 4 ft of perimeter. This is an inventory estimate for light conditions, not a wind rating.

Illustrative estimate

30 bags; fill = $30/75 = 0.40 \text{ yd}^3$; quarter-yard order 0.50 yd^3 . Allow justified return/corner or contingency changes. Student must show the actual selected rule and dimensions.

One team, four teams and deadline staffing

$S = 0.75R$ normally, or $0.50R$ for justified difficult conditions.

One-team time: $30/S$ hours; four-team time: $30/(4S)$ hours.

Adjusted team-hours: $30/S$. Deadline: 2 hours.

Minimum production teams: round up $[30/(S \times \text{usable hours})]$. Delivery delay/route constraints: Adult crews inspect debris and the site. Tarp ballast is not hazardous-material containment.

For illustration only, if $R = 60$ then normal $S = 45$: one team 0.67 h; four teams 0.17 h. Deadline lower bound ignoring setup/delivery is 1 teams. Substitute the student's measured R and actual usable time.

Equipment and actual elapsed time

Start with 2 shovels per production team and one gloves/glasses set per worker, including support workers. Request fill delivery, appropriate moving equipment, closure tools, first aid, water and other materials. Check the route against trial distance.

On a dry level route ≤ 300 ft, each separately staffed wheelbarrow moves at most 24 bags/h and each cart 48 bags/h under catalog assumptions; longer or difficult routes are slower. Round trips up and include operators/loading help. Vehicle access is not guaranteed. Equipment requests must match the chosen number of teams rather than assumed fixed card stock.

Major constraints and judgment

Reasonable for temporary ground-cover edging; sorting and debris disposal remain separate. Tarp, exclusion, labeled sorting zones and removal plan.

Students should identify the likely slowest stage, actual completion time, safe access/drainage, material dimensions, drinking-water delivery, rest/rotation and food for longer work. Accept fewer bags only when a reduced mission is explicit; accept more teams only with equipment, safe workspace and support to use them.

ESTIMATE GUIDE 26 • Potentially Contaminated Runoff

Possible approach, not a prescribed solution

One illustrative narrow three-layer line uses 3 bags/ft. Responders may instead require another containment method; ordinary teams have no authority to enter contamination.

Illustrative estimate

150 bags; fill = $150/75 = 2.00 \text{ yd}^3$; quarter-yard order 2.00 yd^3 . Allow justified return/corner or contingency changes. Student must show the actual selected rule and dimensions.

One team, four teams and deadline staffing

$S = 0.75R$ normally, or $0.50R$ for justified difficult conditions.

One-team time: $150/S$ hours; four-team time: $150/(4S)$ hours.

Adjusted team-hours: $150/S$. Deadline: 1 hours.

Minimum production teams: round up $[150/(S \times \text{usable hours})]$. Delivery delay/route constraints: Planning only: hazardous-material responders authorize all work, containment destinations and PPE. Students never handle contamination.

For illustration only, if $R = 60$ then normal $S = 45$: one team 3.33 h; four teams 0.83 h. Deadline lower bound ignoring setup/delivery is 4 teams. Substitute the student's measured R and actual usable time.

Equipment and actual elapsed time

Start with 2 shovels per production team and one gloves/glasses set per worker, including support workers. Request fill delivery, appropriate moving equipment, closure tools, first aid, water and other materials. Check the route against trial distance.

On a dry level route ≤ 300 ft, each separately staffed wheelbarrow moves at most 24 bags/h and each cart 48 bags/h under catalog assumptions; longer or difficult routes are slower. Round trips up and include operators/loading help. Vehicle access is not guaranteed. Equipment requests must match the chosen number of teams rather than assumed fixed card stock.

Major constraints and judgment

Conditional on hazardous-material responder direction; source control and drain protection may be better.

Responder-specified PPE, compatible liner, exclusion, collection/disposal.

Students should identify the likely slowest stage, actual completion time, safe access/drainage, material dimensions, drinking-water delivery, rest/rotation and food for longer work. Accept fewer bags only when a reduced mission is explicit; accept more teams only with equipment, safe workspace and support to use them.

ESTIMATE GUIDE 27 • Flooded Maintenance Yard

Possible approach, not a prescribed solution

One illustrative broad three-layer line uses 6 bags/ft. Real containment method, compatibility and deployment are responder decisions.

Illustrative estimate

480 bags; fill = $480/75 = 6.40 \text{ yd}^3$; quarter-yard order 6.50 yd^3 . Allow justified return/corner or contingency changes. Student must show the actual selected rule and dimensions.

One team, four teams and deadline staffing

$S = 0.75R$ normally, or $0.50R$ for justified difficult conditions.

One-team time: $480/S$ hours; four-team time: $480/(4S)$ hours.

Adjusted team-hours: $480/S$. Deadline: 4 hours.

Minimum production teams: round up $[480/(S \times \text{usable hours})]$. Delivery delay/route constraints: Water may contain oil or other contaminants. Trained responders control entry, compatible materials, recovery and disposal.

For illustration only, if $R = 60$ then normal $S = 45$: one team 10.67 h; four teams 2.67 h. Deadline lower bound ignoring setup/delivery is 3 teams. Substitute the student's measured R and actual usable time.

Equipment and actual elapsed time

Start with 2 shovels per production team and one gloves/glasses set per worker, including support workers. Request fill delivery, appropriate moving equipment, closure tools, first aid, water and other materials. Check the route against trial distance.

On a dry level route ≤ 300 ft, each separately staffed wheelbarrow moves at most 24 bags/h and each cart 48 bags/h under catalog assumptions; longer or difficult routes are slower. Round trips up and include operators/loading help. Vehicle access is not guaranteed. Equipment requests must match the chosen number of teams rather than assumed fixed card stock.

Major constraints and judgment

Requested bags and fill exceed supply; prioritize isolation, professional recovery and a shorter approved segment. Hazmat coordination, compatible liner, lights, disposal and exclusion.

Students should identify the likely slowest stage, actual completion time, safe access/drainage, material dimensions, drinking-water delivery, rest/rotation and food for longer work. Accept fewer bags only when a reduced mission is explicit; accept more teams only with equipment, safe workspace and support to use them.

ESTIMATE GUIDE 28 • Sand Delivery on a Closed Hill

Possible approach, not a prescribed solution

A narrow two-layer diversion line is one classroom option for shallow runoff. Justify the route and receiving drainage; choose another count if the mission/layout changes.

Illustrative estimate

180 bags; fill = $180/75 = 2.40$ yd³; quarter-yard order 2.50 yd³. Allow justified return/corner or contingency changes. Student must show the actual selected rule and dimensions.

One team, four teams and deadline staffing

$S = 0.75R$ normally, or $0.50R$ for justified difficult conditions.

One-team time: $180/S$ hours; four-team time: $180/(4S)$ hours.

Adjusted team-hours: $180/S$. Deadline: 4 hours.

Minimum production teams: round up $[180/(S \times \text{usable hours})]$. Delivery delay/route constraints: The delivery truck cannot reach the school. The earliest sand drop is 1 hour after assignment; propose a safe last-mile plan.

For illustration only, if $R = 60$ then normal $S = 45$: one team 4.00 h; four teams 1.00 h. Deadline lower bound ignoring setup/delivery is 1 teams. Substitute the student's measured R and actual usable time.

Equipment and actual elapsed time

Start with 2 shovels per production team and one gloves/glasses set per worker, including support workers. Request fill delivery, appropriate moving equipment, closure tools, first aid, water and other materials. Check the route against trial distance.

On a dry level route ≤ 300 ft, each separately staffed wheelbarrow moves at most 24 bags/h and each cart 48 bags/h under catalog assumptions; longer or difficult routes are slower. Round trips up and include operators/loading help. Vehicle access is not guaranteed. Equipment requests must match the chosen number of teams rather than assumed fixed card stock.

Major constraints and judgment

Useful if last-mile transfer can finish; choose a closer approved fill site or mechanized transfer. Delivery plan, sheet, route control, rest and additional transfer capacity.

Students should identify the likely slowest stage, actual completion time, safe access/drainage, material dimensions, drinking-water delivery, rest/rotation and food for longer work. Accept fewer bags only when a reduced mission is explicit; accept more teams only with equipment, safe workspace and support to use them.

ESTIMATE GUIDE 29 • Overnight Supply Depot

Possible approach, not a prescribed solution

A broad three-layer shallow barrier is one classroom option, with returns included or separately estimated. Check forecast depth and leakage. Card 12 is unsafe as a road defense regardless of staffing.

Illustrative estimate

720 bags; fill = $720/75 = 9.60 \text{ yd}^3$; quarter-yard order 9.75 yd^3 . Allow justified return/corner or contingency changes. Student must show the actual selected rule and dimensions.

One team, four teams and deadline staffing

$S = 0.75R$ normally, or $0.50R$ for justified difficult conditions.

One-team time: $720/S$ hours; four-team time: $720/(4S)$ hours.

Adjusted team-hours: $720/S$. Deadline: 12 hours.

Minimum production teams: round up $[720/(S \times \text{usable hours})]$. Delivery delay/route constraints: This may require overnight work. Fill must be delivered; include lights, meals, rotation and verified power.

For illustration only, if $R = 60$ then normal $S = 45$: one team 16.00 h; four teams 4.00 h. Deadline lower bound ignoring setup/delivery is 2 teams. Substitute the student's measured R and actual usable time.

Equipment and actual elapsed time

Start with 2 shovels per production team and one gloves/glasses set per worker, including support workers. Request fill delivery, appropriate moving equipment, closure tools, first aid, water and other materials. Check the route against trial distance.

On a dry level route ≤ 300 ft, each separately staffed wheelbarrow moves at most 24 bags/h and each cart 48 bags/h under catalog assumptions; longer or difficult routes are slower. Round trips up and include operators/loading help. Vehicle access is not guaranteed. Equipment requests must match the chosen number of teams rather than assumed fixed card stock.

Major constraints and judgment

Conditional; rotation, food, lighting and delivery staging are as important as bags. Sheets, powered lights, meals, warm breaks, staged fill and relief workers.

Students should identify the likely slowest stage, actual completion time, safe access/drainage, material dimensions, drinking-water delivery, rest/rotation and food for longer work. Accept fewer bags only when a reduced mission is explicit; accept more teams only with equipment, safe workspace and support to use them.

ESTIMATE GUIDE 30 • Two Sites, One Workforce

Possible approach, not a prescribed solution

A narrow two-layer diversion line is one classroom option for shallow runoff. Justify the route and receiving drainage; choose another count if the mission/layout changes.

Illustrative estimate

420 bags; fill = $420/75 = 5.60 \text{ yd}^3$; quarter-yard order 5.75 yd^3 . Allow justified return/corner or contingency changes. Student must show the actual selected rule and dimensions.

One team, four teams and deadline staffing

$S = 0.75R$ normally, or $0.50R$ for justified difficult conditions.

One-team time: $420/S$ hours; four-team time: $420/(4S)$ hours.

Adjusted team-hours: $420/S$. Deadline: 3 hours.

Minimum production teams: round up $[420/(S \times \text{usable hours})]$. Delivery delay/route constraints: Shelter request is due in 1 hour; recreation request in 3 hours. Protect people and essential access before empty property.

For illustration only, if $R = 60$ then normal $S = 45$: one team 9.33 h; four teams 2.33 h. Deadline lower bound ignoring setup/delivery is 4 teams. Substitute the student's measured R and actual usable time.

Equipment and actual elapsed time

Start with 2 shovels per production team and one gloves/glasses set per worker, including support workers. Request fill delivery, appropriate moving equipment, closure tools, first aid, water and other materials. Check the route against trial distance.

On a dry level route ≤ 300 ft, each separately staffed wheelbarrow moves at most 24 bags/h and each cart 48 bags/h under catalog assumptions; longer or difficult routes are slower. Round trips up and include operators/loading help. Vehicle access is not guaranteed. Equipment requests must match the chosen number of teams rather than assumed fixed card stock.

Major constraints and judgment

Combined plan exceeds bag/fill stock; prioritize shelter, reduce or abandon second request. Priority communication, separate deadlines, accessible shelter entrance and phased allocation.

Students should identify the likely slowest stage, actual completion time, safe access/drainage, material dimensions, drinking-water delivery, rest/rotation and food for longer work. Accept fewer bags only when a reduced mission is explicit; accept more teams only with equipment, safe workspace and support to use them.

TEACHER GUIDE • INCIDENTS & DEBRIEF

Introduce updates after initial plans are complete

Give each student one update, or give the whole class the same one to compare decisions. Read card changes as modifications to their scenario; all other resources remain as stated. Ask students to mark what changes in their sketch, staffing, schedule, supply requests, worker support and priority decision. If an update is irrelevant to a particular structure, use its stated infrastructure adaptation or select another card.

For mid-operation updates, specify time elapsed and completed bags before recalculation. Example: issue the card before work begins, or say “one hour has elapsed and 30 bags are already placed.” Apply factors only to remaining bags; count completed bags once. Do not award credit for a multiplication alone without a revised operational decision.

Teacher revision checks

Delivery delay: shift the critical start or split tasks; no fictitious early fill. Extra volunteers: PPE, shovels, transport and workspace still constrain production. Failed tools/cart: reorganize the chain and staffing. Heat: replace factor, not multiply both, and revise water/rest. Unsafe weather: stop and shelter. New access/drainage restriction: redraw rather than claim an incomplete barrier gives equal protection. Added mission/short supply: prioritize people and essential services, identify residual risk, request alternatives. An impossible request needs an explicit refusal or reduced mission.

Debrief prompts • 10 minutes

What was your measured bottleneck? Did it remain the bottleneck on the new site? Which resource existed but could not be employed? How did a longer route change the apparent production rate? What did a water/rest plan change? Which rejected bag plan was the strongest decision? What uncertainty would you check first as the incident lead? Have each student finish: “Our plan needs ___, delivered by ___, at ___, with ___ people, and will finish by ___ unless ___.”

Optional extensions

Compare two safe carry distances in controlled trials, changing one variable. Use a chart of output versus time to show why a 20-minute rate is not an all-day promise. Allocate a shared truck across three sites with different deadlines. Add 10% bag reserve and recalculate ordered fill. Determine how many extra shovels/carts change the actual bottleneck. Compare basin gallons with approved container delivery, including cleanup and water availability. Estimate uncertainty ranges instead of inventing precise engineering strength.

TEACHER BACKGROUND • SOURCES & LIMITS

Sources checked for this package

U.S. Army Corps of Engineers, Jacksonville District, Sandbag Information:

<https://www.saj.usace.army.mil/Missions/Emergency-Operations/Sandbag-information/>

Background for partial filling, gloves, staggered placement and plastic use. The classroom trial's consistent closure/counting standard is an instructional choice.

U.S. Army Corps of Engineers, Rock Island District, NFFMC Supplies:

<https://www.mvr.usace.army.mil/missions/emergency-management/nffmc/supplies/>

Background for temporary barriers, sheet ballast and plastic sheeting; not validation of the classroom count geometry.

National Weather Service, Flood Watch vs. Warning:

<https://www.weather.gov/safety/flood-watch-warning>

Background for imminent flood hazards and avoiding flooded roads. Check official local forecasts and school procedures on the day of the actual outdoor lesson.

National Weather Service Pittsburgh:

<https://www.weather.gov/pbz/>

Local weather information; scenarios are fictional, not current forecasts or site assessments.

Authorship of simplified assumptions

The universal 75 bags $\approx$ 1 yd³ conversion comes from the course brief and is used consistently. One-foot bag length, three-inch modeled layers, structural bag counts, catalog trip capacities/cycles, 0.75/0.50 rate factors, water stock and food intervals are supplied classroom assumptions. They are not quoted agency design standards or medically prescribed doses. Water conversion: 1 ft³ $\approx$ 7.48 gallons; approximate water load uses 8.3 lb per gallon. Do not extend water-load math to bag weighing.

Limits to make explicit

A filled bag must be moved and positioned to become useful. Barriers leak; drainage destinations and neighbors matter. Simplified structures do not establish a safe flood height, slope strength, retaining capacity, containment compatibility or equipment anchorage. Deep or moving floodwater, unstable ground, electricity and hazardous materials require qualified response and may require withdrawal. Students plan on paper; their field task remains the safe, supervised production experiment.